Household Sound Detection

Technical evidence summary, October 2026: methods, results and limitations.

Summary. On a 15-sound household benchmark with simulated noise, room echo and very short clips, our fine-tuned model detects **56%** of real events at a 1% false-alarm rate, against **43%** for Apple's built-in sound classifier. It retains **54%** after compression to 8-bit precision, small enough for low-cost embedded chips. On clean studio-quality audio, Apple's classifier is slightly ahead. All results to date use public recordings; real-home validation is the next step.

1. What we have built

- **Smoke and CO alarm detector.** Identifies the standard alarm beep rhythm (ISO 8201: three beeps for smoke, four for carbon monoxide) rather than relying on a sound label. Implemented in Swift for iPhone and in C for embedded chips, with identical results on shared test files. It detected 3 of 12 real fire-alarm recordings; the other 9 are bells, horns or 2-second snippets with no beep pattern, which remain a gap.
- **On-phone model.** A 17 KB model for glass breaking, knock, doorbell and baby crying, running alongside Apple's classifier inside the app with a per-sound rule for how many consecutive sound windows must agree before an alert.
- **15-sound networks.** Public EfficientAT networks (MIT licence) fine-tuned end to end on commercially licensed audio. A small network (0.72 million weights, 0.72 MB at 8-bit) and a large one (4.2 million weights, 4.2 MB at 8-bit). Sounds: alarm, baby crying, cough, dog bark, door slam, doorbell, glass breaking, gunshot, knock, microwave, phone ring, scream, siren, toilet flush and running water.
- **Opt-in clip review.** A "Sounds heard" page keeps a 5-second clip of each alert on the user's phone only, so the user can mark it Right or Wrong. This is the mechanism for collecting real-home evaluation data.

2. Main result: hard-conditions benchmark

Each sound is scored in six conditions built from held-out FSD50K evaluation clips: clean, noise at 10 dB, noise at 0 dB, room echo, 1-second blips, and noise, echo and short clips combined. Noise and echo are simulated to approximate home conditions. The metric is the share of real events detected when only 1% of non-events may trigger a false alarm, so no model benefits from a favourably chosen alert threshold.

Model (mean of 15 sounds, six conditions)	Detected at 1% false alarms	Ranking quality (AUC, 1.0 = perfect)
Apple built-in classifier	43.3%	0.776
Ours, small, full precision	49.4%	0.925
Ours, small, 8-bit	47.3%	0.922
Ours, large, full precision	55.6%	0.936
Ours, large, 8-bit	53.9%	0.933

Large model by condition: clean 73%, noise 10 dB 61%, echo 64%, 1-second blip 51%, noise 0 dB 42%, all combined 41%.

3. Results by sound: Apple against our large 8-bit model

Detected at 1% false alarms / AUC, averaged over the six conditions. Our model is ahead on 11 sounds (two of them weak for both systems), level on 3, and behind on 1. The small 8-bit model is weaker on baby crying (.60), siren (.62) and doorbell (.24).

Sound	Apple	Ours (large, 8-bit)	Result
Glass breaking	.39 / .71	.66 / .96	Clearly ahead
Knock	.40 / .75	.71 / .95	Clearly ahead
Scream	.27 / .62	.52 / .96	Clearly ahead
Gunshot	.34 / .60	.49 / .93	Clearly ahead
Cough	.58 / .84	.71 / .96	Ahead
Toilet flush	.69 / .97	.81 / .99	Ahead
Running water	.46 / .88	.55 / .96	Ahead
Dog bark	.70 / .87	.75 / .96	Slightly ahead
Phone ring	.24 / .63	.32 / .92	Ahead (weak for both)
Microwave	.38 / .76	.42 / .91	Slightly ahead
Door slam	.11 / .64	.25 / .89	Ahead (weak for both)
Alarm (generic)	.09 / .69	.14 / .83	Weak for both; smoke and CO are covered by the beep-pattern detector
Baby crying	.75 / .96	.72 / .97	Level
Doorbell	.33 / .77	.32 / .88	Level (26 test clips)
Siren	.77 / .93	.72 / .95	Apple ahead

4. Independent check: ESC-50, clean recordings

ESC-50 is a standard 2,000-clip benchmark that we did not train on. Seven sounds overlap with ours (dog, cough, toilet, siren, glass, knock, baby crying); the negatives are the other 1,720 clips.

Model	Detected at 1% false alarms	AUC
Apple built-in classifier	98.6%	0.998
Ours, large, full precision	96.1%	0.998
Public EfficientAT mn20 as released (no training by us)	93.2%	0.996
Ours, small, 8-bit	90.7%	0.994

On clean, standard data every model is near the ceiling and Apple's classifier is marginally best. Our advantage is in degraded conditions: noise, echo, faint and very short sounds.

5. Realistic stream: on-phone model against Apple

A 30-minute simulated stream with 36 instances of each sound mixed into household background. Events found out of 36 / false alarms per hour:

Sound	Apple	Our on-phone model
Glass breaking	9 / 0	16 / 6
Baby crying	13 / 0	20 / 6
Doorbell	1 / 0	8 / 0
Knock	12 / 2	12 / 2

Our model finds more events; Apple raises fewer false alarms. Reducing glass false alarms (currently 6 per hour) is an open item.

6. Competitive landscape

Competing products run detection inside their own devices and do not publish accuracy, so they cannot be scored on our benchmark. The comparison below is of coverage, taken from each company's own documentation.

Product	Sounds detected (as stated by the maker)	Notes
Amazon Alexa Guard (Echo)	Smoke alarm, CO alarm, breaking glass; later also human activity	Launched 2019; user switches it on when leaving. Current availability unconfirmed. TechCrunch, Security Sales
Google Home speakers and displays	Smoke alarm, CO alarm, breaking glass (US only, Google Home Premium)	Google Home Premium help
Google Nest cameras	Dog bark, speech, smoke and CO alarms; user selects which sounds alert	Certain camera models only. Google help
Wyze Cam	Smoke (T3) and CO (T4) beep patterns	Same pattern-based approach as ours; user reports of missed newer alarms. Wyze help
Seeed Sound Event Detection Module D1	Baby cry, glass break, gunshot, T3/T4 alarms, snoring	On-device chip module. Maker claims 95% or better recognition; not independently tested. Product page
Apple Sound Recognition	About 300 labels in the developer classifier	The competitor we measured directly. Apple documentation
Ours	15 sounds plus smoke and CO beep patterns	Measured in sections 2 to 5.

Alexa and Google's speakers each cover three sounds; our system covers fifteen. Coverage does not establish accuracy on the shared sounds, which no one outside those companies has measured. A direct device-level comparison, playing identical clips to each product and counting alerts, is planned.

7. Comparison with a general-purpose audio language model

We also tested OpenAI's audio model (gpt-audio-1.5), which takes real audio, exactly as released and with no examples. Both systems were given the same 500 clips (10 clean and 10 hard recordings of each of the 15 sounds, plus 100 clean and 100 hard background clips) and had to pick one of 16 labels. Chance is not 1 in 16 here: answering "background" every time scores 40%.

System	All 500 clips	Sounds, clean	Sounds, hard conditions	Background
Ours (large, 8-bit)	59%	81%	54%	46%
OpenAI gpt-audio-1.5	48%	67%	29%	48%

The audio model is respectable on clean recordings (very good on baby crying and running water) but falls away in hard conditions, and it found 2 of 20 alarms and 5 of 20 gunshots. Nineteen of its answers were not a usable label and are counted as wrong.

Read with care: this is a different, simpler measure from sections 2 to 5 (one label per clip, not a false-alarm-controlled alert), our model's answer is its highest-scoring label averaged over the clip, and our model was trained on this kind of audio while the language model was not. The result shows what a general-purpose model achieves with no tuning, not a ceiling for such models. A cloud audio model would also need continuous home audio sent off the device.

8. Limitations

- **Real-home validation is pending.** Noise and echo are simulated. Training and test clips both come from FSD50K, as separate sets, so the hard-conditions benchmark is more favourable to us than a fully independent one; ESC-50 (section 4) is the independent check.
- **Comparison scope.** We measured Apple's classifier, one untrained public network and one general-purpose audio language model. Stronger public models (BEATs, PaSST) and other audio language models are untested, and we do not claim state of the art.
- **Alert settings are conservative.** At up to one false alarm per hour, calibrated settings detect roughly 30 to 50% of events in a crowded stream for the stronger sounds (glass 12 of 32, dog 21 of 32), and few of door slam, doorbell, generic alarm and scream.
- **Embedded hardware.** Model sizes and operation counts are calculated; they have not yet been measured on a target chip.
- **Data licensing.** Training uses only CC0 and CC BY clips and Donate-a-Cry (ODbL). ESC-50 (CC BY-NC) is used for testing only. Attribution will be included at release.

9. Next steps

- Collect real-home clips through the opt-in Sounds heard feature and rerun this benchmark on them.
- Learn each home's own sounds (doorbell, washing machine, kettle) from signals the user already gives, such as opening the door after a doorbell alert.
- Evaluate larger public networks and other audio language models (with example clips), run the compressed model on the target chip, and compare directly against competing devices.

10. Sources

- EfficientAT networks (MIT licence): github.com/fschmid56/EfficientAT; paper: arxiv.org/abs/2211.04772

- FSD50K dataset: zenodo.org/records/4060432; paper: arxiv.org/abs/2010.00475
- ESC-50 benchmark: github.com/karolpiczak/ESC-50
- Donate-a-Cry corpus: github.com/gveres/donateacry-corpus
- Apple SoundAnalysis: SNClassifySoundRequest documentation

Full result files, test code and reproduction steps are available on request.